

READ FIRST - COMP2000 Traffic Simulation

Adapted from Yakov's Codebase Reading and Defence Guide (Sept 2026), split per pair by Henry. Full PDF: [docs/COMP2000_Traffic_Simulation_Codebase_Guide.pdf](#).

You don't need to read the whole codebase. Read this file, then open **your pair's** guide only. Cross-read into other pairs' guides when your code touches theirs.

The project in one minute

Java Swing traffic simulation. Cars, trucks, buses and emergency vehicles travel straight through a grid of signalised intersections. No turns, no roundabouts, no pedestrians, no physics engine.

Each tick: update lights -> read sensors -> move vehicles -> despawn off-map -> maybe spawn -> repaint.

```
network file -> roads, lanes, intersections, bus stops
                |
                v
each tick: update lights -> read sensors -> move vehicles
                        -> despawn -> maybe spawn -> repaint
```

Pair assignments

Pair	Members	Owns	Read this guide
A	Ali, Henry, Oscar	Engine, sensors, spawning, observer, display	pair-a-engine.md
B	Callum, Jubril	RoadNetwork, Road, Lane, BusStop, NetworkLoader	pair-b-network.md
C	Ben, Jacob	Vehicle + Car/Truck/Bus/EmergencyVehicle, DriverProfile	pair-c-vehicles.md
D	Addrita, Cam	TrafficLight + states, Intersection, SignalisedIntersection	pair-d-lights.md

How to run

From the repo root (`C:\Users\<you>\COMP2000` or wherever you cloned it):

PowerShell (Windows):

```
javac -d src/out $(Get-ChildItem src/trafficsim -Recurse -Filter *.java | ForEach-Object { $_.FullName })
java -cp src/out trafficsim.Main
```

Bash / Git Bash:

```
javac -d src/out $(find src/trafficsim -name "*.java")
java -cp src/out trafficsim.Main
```

Optional args: - <network-file> - path to a network definition (defaults to networks/grid.txt) - --seed=<long> - repeatable RNG seed

There is no start button. The moment the window opens the sim is running. Bottom bar has Pause / Resume / Step / Reset / Spawn 1. Space toggles pause, **s** steps once, **n** spawns one, **r** resets.

Defence checklist (everyone)

- Run the program once normally and once with `--seed=123`.
- Point at the dashboard seed and explain reproducibility.
- Follow one car through `SimulationEngine` -> `TrafficSensor` -> `Vehicle.move`.
- Explain the centre-distance-to-bumper-gap calculation in your own words.
- Show a red light, then explain `STOP_LINE_RADIUS` and the all-red phase.
- Show a bus stop and explain why a bus only serves it when its lane matches.
- Show an emergency vehicle and state: **priority, not collision immunity**.
- Know that the display *reads* state; it does not control traffic movement.
- Be honest about scope: no turns, roundabouts, pedestrians or physics engine.

Constants everyone should recognise

Constant	Value	Meaning
LANE_WIDTH / LANE_HALF_WIDTH	18 / 9	Road is 36 units of asphalt; lane centres sit 9 units from road centre.
STOP_LINE_RADIUS	52	Virtual stop line distance before intersection centre.
SIGHT_RANGE	250	Hazards farther than this are ignored per tick.
MIN BUMPER_GAP	4	Free space after subtracting vehicle half-lengths.
SPAWN_CAP	18	Max active vehicles before spawning pauses.
SPAWN_PROBABILITY	0.06	Chance per tick of attempting a spawn.
DWELL_TICKS	30	How long a bus waits at a served stop.
ADAPTIVE_QUEUE_HIGH	4	Vehicles queued before green may extend.

For every class you touch, ask three questions

1. What state does it own?
2. What does it change?
3. Who calls it?